

Undergraduate Common Entrance Examination for Design

UCEED 2018

Paper1

**UCEED 2018
COMPLETE
SOLUTION**

Sai Praveen

Paper Specific Instructions

1. The examination is of 3 hours duration. There are a total of **85 questions** carrying a total of 300 marks. The paper is divided into three sections, **A, B** and **C**. All sections are compulsory. Questions in each section are of different types.
2. **Section A (80 Marks)** contains a total of 20 **Numerical Answer Type (NAT)** questions. For each question, the answer is a real number that needs to be entered using the virtual keyboard on the monitor. No choices will be shown for these questions. There is NO NEGATIVE marking for this section. Each correct answer will be awarded 4 marks. Questions not attempted or answered incorrectly will be given zero marks. Questions from 1 to 20 belong to this section.
3. **Section B (100 Marks)** contains a total of 25 **Multiple Select Questions (MSQ)**. Each question may have **one or more than one correct choice(s)** out of the four given. A candidate gets 4 marks only if ALL the correct choices and NO wrong choices are selected for each question. There is NO PARTIAL marking and NO NEGATIVE marking for this section. Questions not attempted or with incorrect or incomplete answers will be given zero marks. Questions from 21 to 45 belong to this section.
4. **Section C (120 Marks)** contains a total of 40 **Multiple Choice Questions (MCQ)**. Each question has four choices out of which ONLY ONE is the correct answer. There is NEGATIVE marking for this section. Each correct answer will be awarded 3 marks and each wrong answer will receive –1 (minus 1) mark. Questions not attempted will be given zero marks. Questions from 46 to 85 belong to this section.
5. Calculators, charts, graph sheets, tables, cellular phone, smart watches and/or other electronic gadgets are **NOT** allowed in the examination hall.
6. Papers will be provided for rough work.

Section A: Numerical Answer Type Questions

Section A (80 Marks) contains a total of 20 **Numerical Answer Type (NAT)** questions. For each question, the answer is a real number that needs to be entered using the virtual keyboard on the monitor. No choices will be shown for these questions. There is **NO NEGATIVE** marking for this section. Each correct answer will be awarded 4 marks. Questions not attempted or answered incorrectly will be given zero marks. Questions from 1 to 20 belong to this section.

Q.01 How many different types of characters appear in the figure given below?



Ans 21

Q.02 Given below is a series of numbers. Which number will replace the question mark?

101, 103, 107, 109, ?, 127, 131, 137, 139, 149, 151, 157, 163

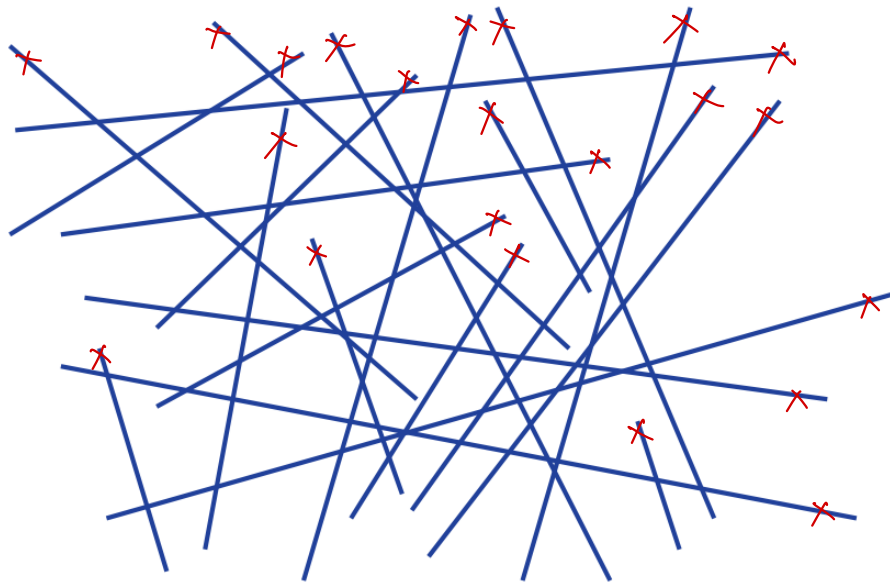
Ans 113

↓
113 (prime numbers)
The following series is prime numbers

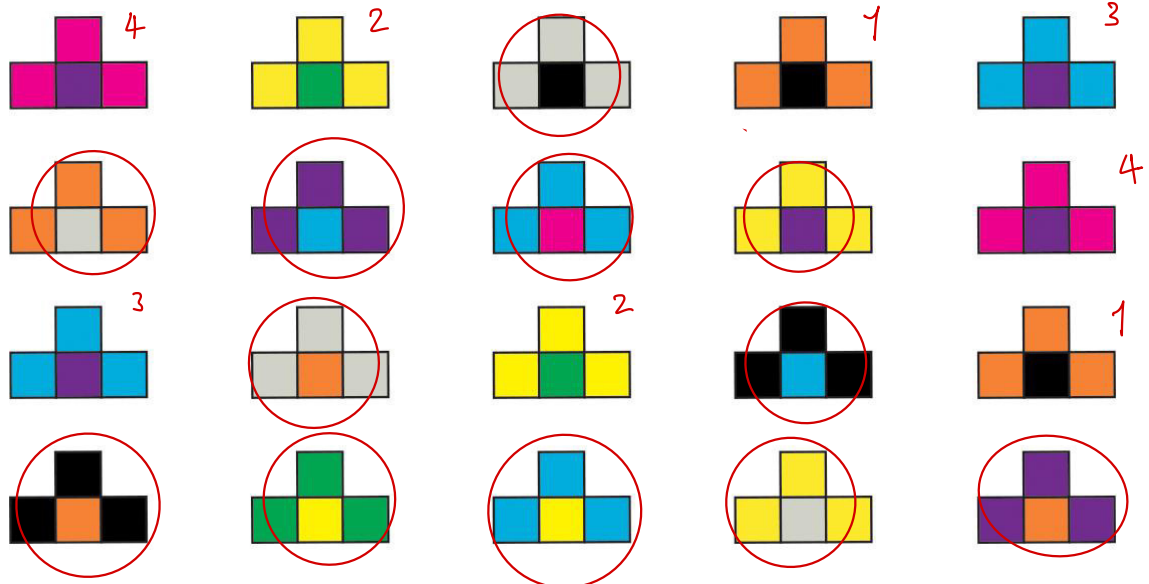
Q.03 How many lines are there in the image given below?

Ans.

22



Q.04 How many configurations of blocks appear only once in the figure given below?

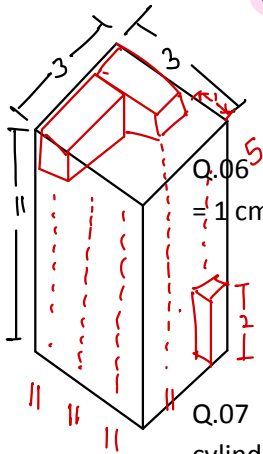


Ans. 12

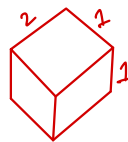
Q.05 What is the number of fonts used in the words given below?

1 2 3
4 5
6
Ans 6

Mango Papaya Apple Banana
Pineapple Orange Grapes Pear
Guava Tomato Pomegranate



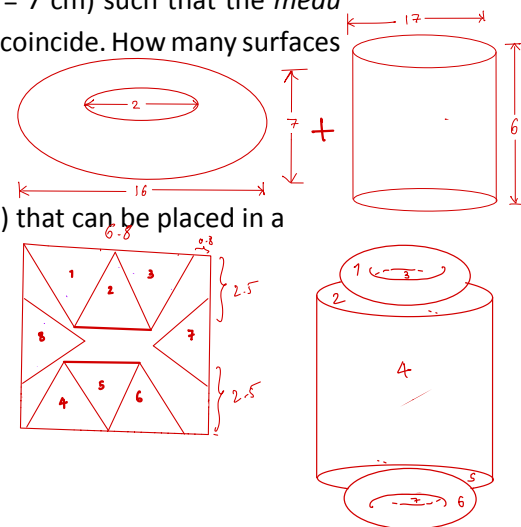
Q.06 What is the maximum number of whole cuboids of length = 2 cm, breadth = 1 cm and height = 1 cm that can be packed into a larger cuboid of length = 3 cm, breadth = 3 cm and height = 11 cm?



$$\therefore 11 + 11 + 11 + 11 + 5 = 49$$

Q.07 Prof. Paithiyam, in an attempt to create the world's most delicious dish, managed to merge a cylindrical *Idli* (diameter = 17 cm, height = 6 cm) with a torus shaped *medu wada* (inner hole diameter = 2 cm, outer diameter = 16 cm, and tube diameter i.e height of torus = 7 cm) such that the *medu wada* is parallel to the base of the *Idli* and the centroids of the two solids coincide. How many surfaces does the resulting solid contain?

Ans 7



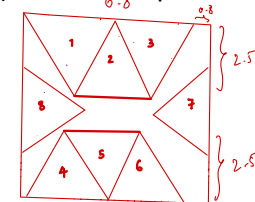
Q.08 What is the maximum number of equilateral triangles (side 3 cm) that can be placed in a square of side 6.8 cm without overlapping each other?

$$h = \frac{\sqrt{3}}{2} a$$

$$h = \frac{3\sqrt{3}}{2}$$

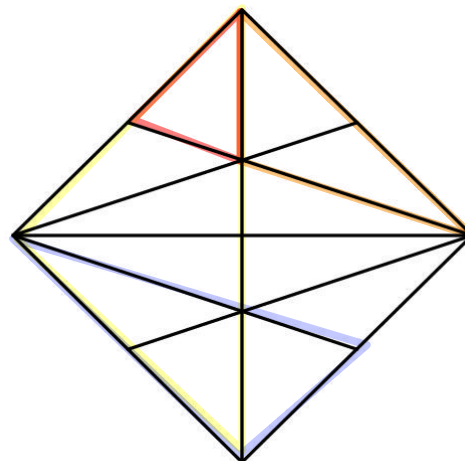
$$= 1.5 \times 1.7$$

$$= 2.55$$

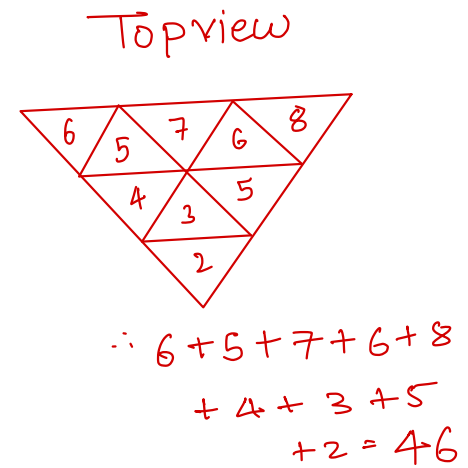
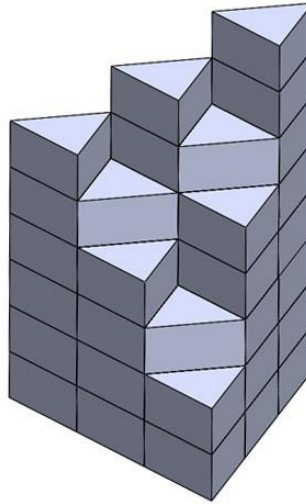


Q.09 How many triangles are there in the figure shown below?

$$\begin{array}{r} 12 \\ 12 \\ 8 \\ + 8 \\ \hline 40 \end{array}$$

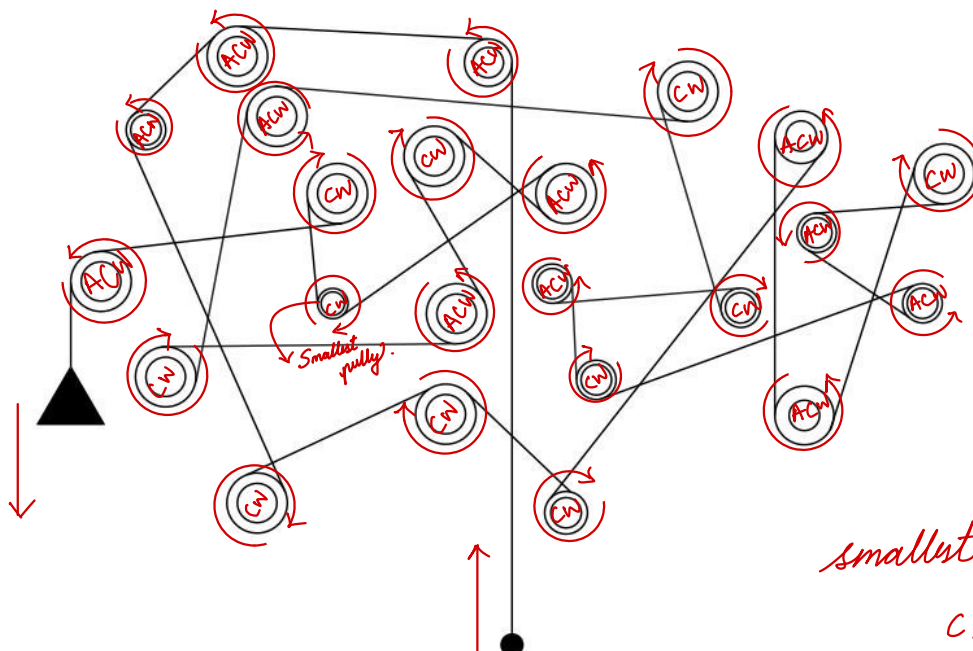


Q.10 Count the number of stacked triangular pieces in the image shown below.



Ans 46

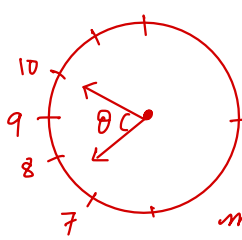
Q.11 Shown below is a combination of fixed pulleys. If the triangular load moves down, how many pulleys will rotate in the same direction as the smallest pulley (including itself)? The overlapping paths do not interfere with each other.



smallest pulley moves
 CW $\therefore$ total pulleys
 are 11

Ans 11

current time = 3:10
after 4 hrs 40 min, time is = 7:50



$\theta = ?$
 $\therefore$ hour hand has covered 210° but when minute hand is at '10' then hour will somewhere '7' and '8' and minute hand has moved 300°

Q.12 The current time is 10 minutes past 3 in the afternoon. What will the angle be between the hour hand and the minute hand of the clock, 4 hours and 40 minutes from now?

$$\begin{aligned} 1 \text{ hr} &= 60 \text{ min} \\ 1 \text{ hr} &= 30^\circ \\ 60 \text{ min} &= 30^\circ \\ \therefore 1 \text{ min} &= \frac{1}{2}^\circ \end{aligned}$$

$$\begin{aligned} \therefore \text{hr hand} & \quad \therefore x^\circ = 25^\circ \\ 1 \text{ min} & \xrightarrow{\text{moves}} \frac{1}{2}^\circ \\ 50 \text{ min} & \rightarrow x^\circ = 300 - (210 + 25) = 65^\circ \end{aligned}$$

Q.13 The word 'Ape' is an anagram of 'Pea'. How many of the following words are anagrams of animals?

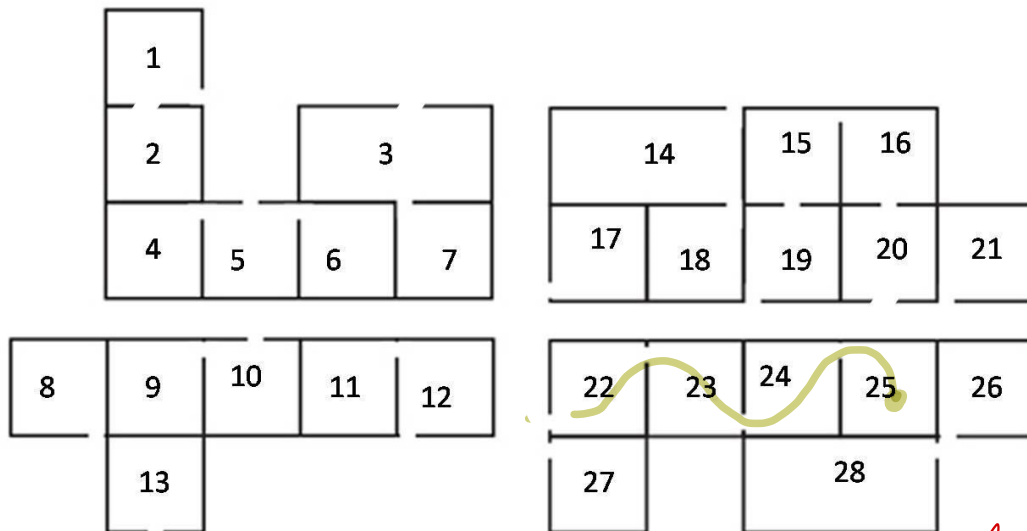
$\rightarrow$ a word (or) phrase that is made by arranging the letters of another word in a different order.

Glow, flow, loin, sit, balm, bare, god, lane, act, tab, bit, tar, reef

wolf lion lamb bear dog cat bat rat

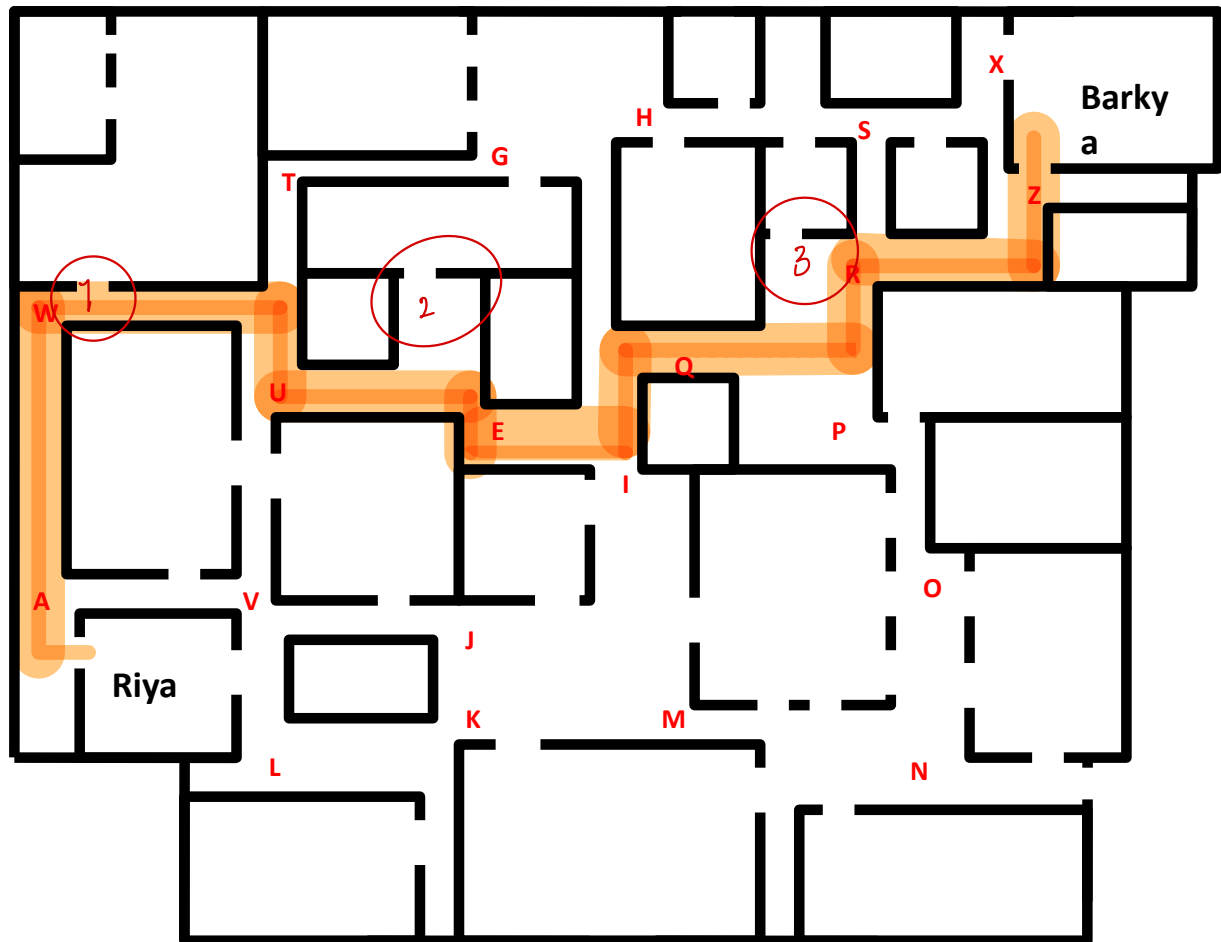
Ans. 8

Q.14 Privacy of a space depends on the level of closure it has from its surroundings i.e. the difficulty of accessing that space from outside the house. The figure below shows the plans of four houses at a village street junction. Which is the most private room amongst all the rooms across all the houses?

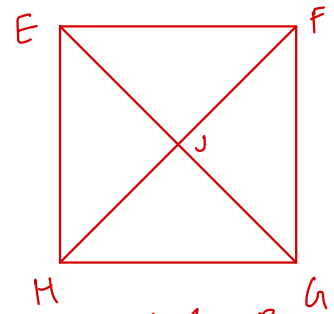
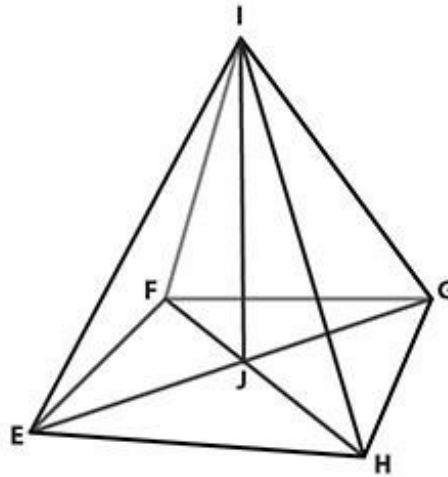


Ans 25

Q.15 The figure below shows the layout plan of the conservative Themmalwadi village which consists of a cluster of houses. Black lines represent walls and the openings in those walls represent either windows or doors. Also marked are the houses of Riya (who lives with her family) and Barkya, who are in love and having an affair. So, hidden from the eyes of the villagers, they regularly meet at night in Barkya's house. Themmalwadi is so conservative that even the sight of women on the street at night raises eyebrows. If Riya takes the path that has the least number of openings, what is the number of openings that she will have to pass by to get to Barkya's house?



Q.16 Shown in the figure is a pyramid with a square base, formed by vertices EFGH and a top vertex I. The diagonally opposite vertices, EG and FH are connected to form the vertex J at their intersection. The vertices I and J are also connected as shown. How many triangles are formed by all these vertices?



no. of Δ s = 8

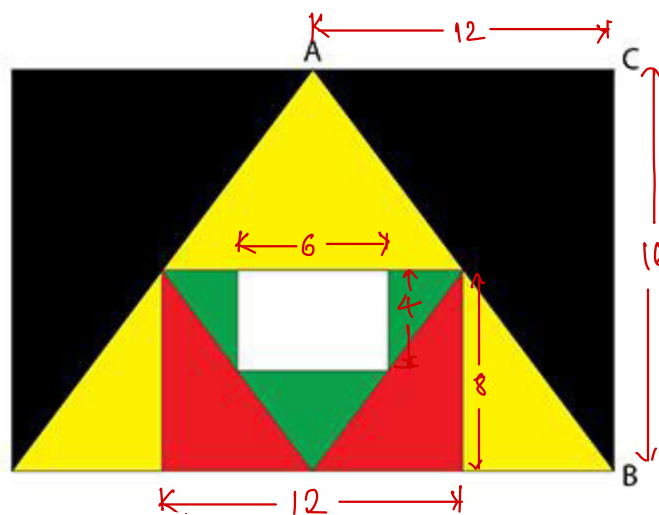
$IHG, IGF, IFE, IEH = 4$

$IJE, IJH, IJG, IJF = 4$

$IEG, IHF = 2$

$\therefore$ Total Δ s = $8 + 4 + 4 + 2$
= 18

Q.17 The length and breadth of the white rectangle are 6 and 4 respectively. The proportions of the outer black, middle red and inner white rectangles are the same. What is the length of the hypotenuse of triangle ABC?



$$\begin{aligned} \therefore AB^2 &= 12^2 + 16^2 \\ &= 144 + 256 \\ &= 400 \\ AB &= \sqrt{400} = 20 \end{aligned}$$

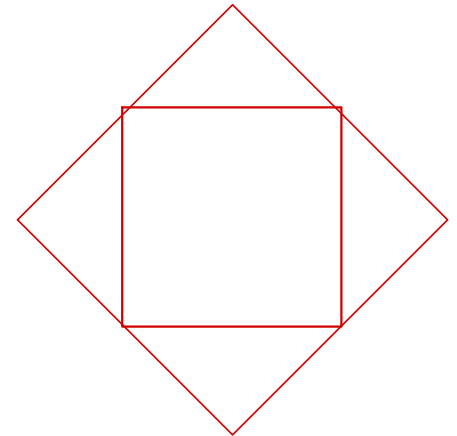
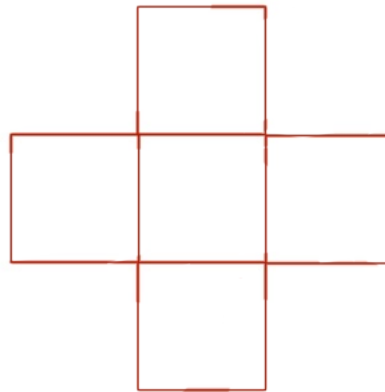
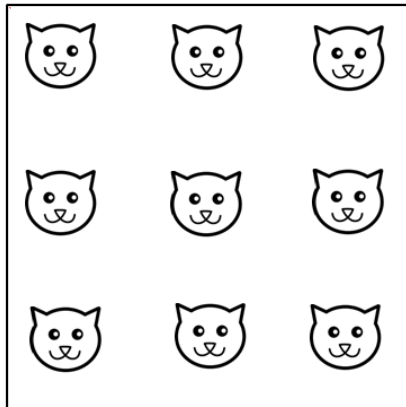
Mukund
2T = 1L
3IC = 1T

Ranjana
1IC = 2L
3T = 1L

Q.18 Meethapur works on a barter system. Each person there loves different sweets. Mukund is willing to exchange 2 toffees for a lollipop and 3 ice creams for a toffee. Ranjana is willing to exchange one ice cream for 2 lollipops and 3 toffees for a lollipop. Mukund and Ranjana don't know each other, but both of them know Sita who loves toffees. Sita has one lollipop. She can meet Ranjana and Mukund only once (not necessarily in that order). What is the maximum number of toffees that Sita can get?

Sita has 1L
 $\therefore$ 1st Sita will go to mukund and exchange her lollipop with 2T and again with that 2T she exchange with 6IC and then she goes to Ranjana and exchange 6IC with 12L and then with that 12L she again enchanges to get $12 \times 3 = 36T$

Q.19 Atul has nine cats that always fight with each other. He puts all of them into a square box. What is the least number of square partitions he must use to keep all the cats separated from each other? You may use any size of square partitions.



4 and 2

Q.20 How many differences are there in the two images shown below?

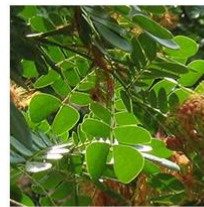
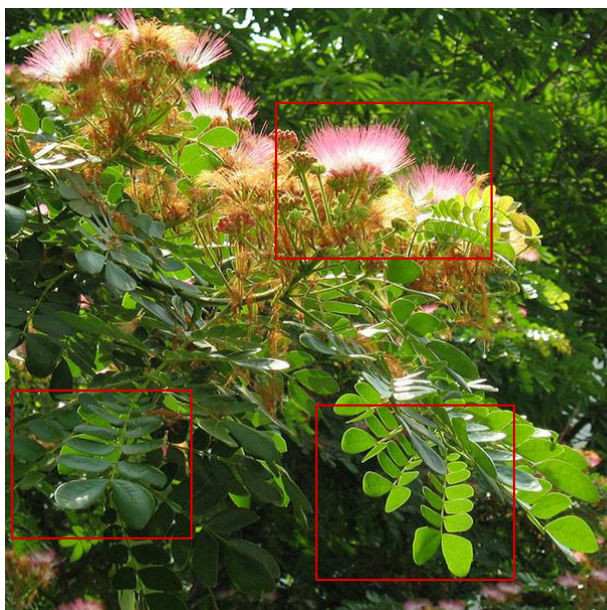


Ans 10

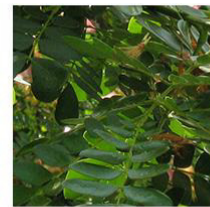
Section B: Multiple Select Questions

Section B (100 Marks) contains a total of 25 **Multiple Select Questions (MSQ)**. Each question may have **one or more than one correct choice(s)** out of the four given. A candidate gets 4 marks only if **ALL** the correct choices and **NO** wrong choices are selected for each question. There is **NO PARTIAL** marking and **NO NEGATIVE** marking for this section. Questions not attempted or with incorrect or incomplete answers will be given zero marks. Questions from 21 to 45 belong to this section.

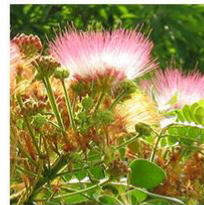
Q.21 On the left is a photograph. Which option(s) is/are part(s) of that photograph?



A



~~B~~



~~C~~



~~D~~

Q.22 If spaces and special characters are ignored in the following text, which country/countries is/are hidden in the text shown below?

I think the earth's ecosystem will collapse very soon. Natural resources have been exploited for far too long at the cost of the wellbeing of humankind, I am afraid. I don't want to stir another controversy but I ardently feel that the time has come for us to take urgent actions to save our planet from manmade disasters.

A. Ghana

~~B.~~ India

C. Peru

~~D.~~ Iran

Q.23 The CBI is investigating a dangerous gang of three criminals. The criminals are Donga, Tirudan and Daaku. The investigation team consists of Pradyuman, Daya, Fredricks, Salunkhe, Tarika and Purvi. Each criminal is to be investigated by exactly two investigators, and each investigator investigates exactly one criminal. The investigation is to be conducted subject to the following constraints:

Pradyuman and Daya do not investigate the same criminal.

Fredricks and Salunkhe investigate the same criminal.

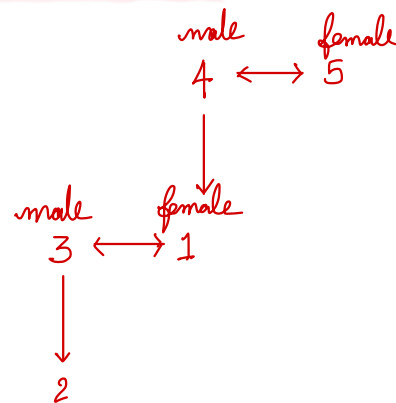
If Fredricks investigates Tirudan, then Tarika must investigate Daaku.

Given these constraints, which of the following could be the acceptable assignment(s) of investigators to potential criminals?

- A. Fredricks & Salunkhe investigate Tirudan, Pradyuman & Daya investigate Donga and Tarika & Purvi investigate Daaku
- ☒ B. Fredricks & Salunkhe investigate Donga, Tarika & Pradyuman investigate Tirudan and Daya & Purvi investigate Daaku
- C. Tarika & Purvi investigate Donga, Fredricks & Pradyuman investigate Tirudan and Salunkhe & Daya investigate Daaku
- ☒ D. Daya & Purvi investigate Tirudan, Tarika & Pradyuman investigate Daaku and Fredricks & Salunkhe investigate Donga

Q.24 If, $9 + 8$ means 9 is the child of 8, and $7 * 6$ means that 6 is the wife of 7, then which of the following statements can be derived from $2 + 3 * 1 + 4 * 5$?

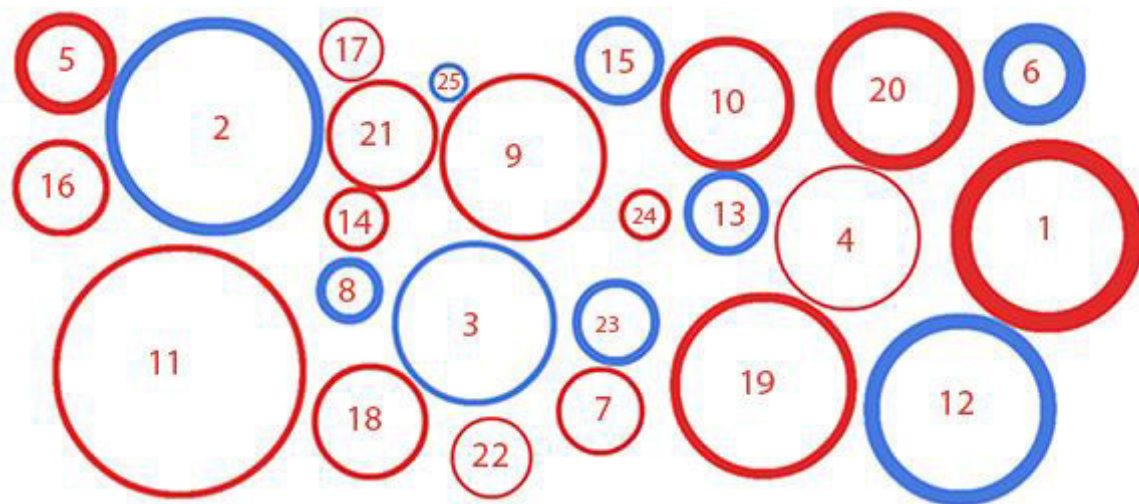
- ☒ A. 2 is the grandchild of 4
- ☒ B. 1 is the daughter of 5
- C. 5 is the grandfather of 3
- ☒ D. 1 is the mother of 2



Q.25 30 circles are given below, with a set of rules:

1. The bigger the circle, the farther away it is from us.
2. The thinner the line of the circle, the closer it is to us.
3. The red circles are farther away from us as compared to the blue circles.

Which of the options given below is/are TRUE?



1. The bigger the circle, the farther away it is from us.
 2. The thinner the line of the circle, the closer it is to us.
 3. The red circles are farther away from us as compared to the blue circles.
- Which of the options given below is/are TRUE?

- ☒ A) Circle 25 is the closest to us
- ☐ B) Circle 25 is farthest from us
- ☒ C) Circle 24 is closer to us than circle 14
- ☐ D) Circle 6 is closer to us than circle 8

Q.26 Which concept does this image by Escher illustrate?



- ☒ A. Tessellation
- ☒ B. Figure and Ground
- ☐ C. Mise en Scene
- ☐ D. Collage

→ It is created when a shape is repeated over and over again covering a plane without any gaps (or) overlap.

→ people instinctively perceive object as either foreground (or) background.

grotesque: distorted and unnatural in shape or size.

Q.27 Read the passage given below, written by Nick Hughes:

"We tend to treat power as though it is intrinsically valuable. We seek it out and covet it, quite irrespective of how we might wield it and what it might get us. One need only look at the history of totalitarian politics to recognise this tendency in its most grotesque form. But power isn't intrinsically valuable, it's only instrumentally valuable – valuable as a means to an end. And whether or not they are objectively valuable, the ends that matter to us, the things that we care about most – our relationships, our projects and goals, our shared experiences, social justice, the pursuit of knowledge, the creation and appreciation of art, music and literature, and the future and fate of ours and other species – do not depend to any considerable extent on our having control over a vast but largely irrelevant Universe. We might be distinctly lacking in power from the cosmic perspective, and so, in a sense, insignificant. But having such power and such significance wouldn't make much of a difference anyway. To lament its lack and respond with despair and nihilism is merely a form of narcissism. Most of what matters to us is right here on Earth."

characterized by gov. in which the political authority exercises absolute and centralized.

with respect to its inherent nature

admiration for yourself

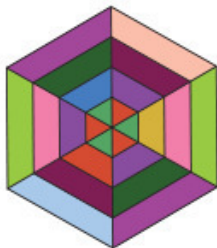
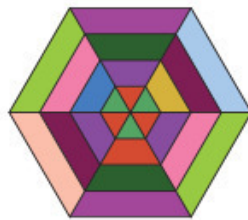
↳ The belief that things do not exist. sense that every thing is unreal.

Which of the statements is/are TRUE of the paragraph above:

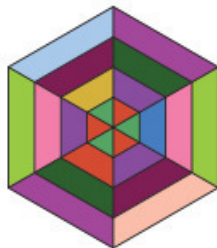
- ☒ A. The author argues that intrinsic value is distinct from instrumental value and that power possesses the latter of the two.
- ☒ B. "Humans and our significance in the universe", is one of the themes discussed in the passage.
- ☐ C. The philosophies of nihilism and narcissism are suggested as remedies by the author.
- ☐ D. The author suggests that we look at the history of totalitarian politics to recognise that we shun power in the most grotesque form.

↳ avoid and stay away from deliberately.

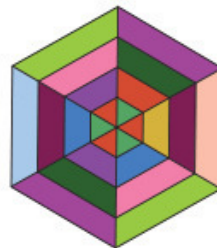
Q.28 Which option(s) is/are simple rotations of the figure given below?



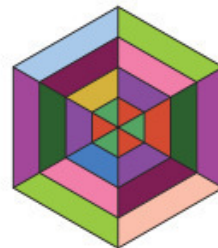
~~A~~



~~B~~



~~C~~



~~D~~

Q.29 Read the statements below:

All female and male natives in their country are strong.

Some men in their country are migrants.

All migrants in their country are not strong.

Not all migrants are women.

Based on the above, which conclusion(s) stated in the options is/are valid?

- ☒ A. Some women in their country are strong.
- ☒ B. Some migrants in their country are men.
- ☒ C. Some people in their country are not strong.
- ☐ D. All strong people in their country are women.

Q.30 Which of the following words are similar in meaning?

- ☒ A. Scribble → *An aim less drawing.*
- ☒ B. Doodle → *An aim less drawing.*
- ☐ C. Nibble → *bite off very small pieces*
- ☐ D. Feeble → *Lacking strength*

Q.31 Which of the following words is/are similar in meaning to 'Centre'?

- ☒ A. Heart
- ☒ B. Core
- ☒ C. Nucleus
- ☒ D. Nub

Q.32 Consider this poem:

Now, marriages happen in halls
They used to happen at home
Honeymoon too
Before they wandered into halls
Stage. Actors. Audiences from afar.
Receptions never really took place
Yet they would go on for days.
Now, receptions include

Those who don't really figure.
Show face, shove envelope (gifts not accepted)
Grin for the camera.
Get out. Not all that bad.
Quick and efficient. No one really cares.

Which mood(s) does the poem evoke?

- ☒ A. Cynicism → *The feeling of distrust or that something isn't going to work out well.*
- ☒ B. Nostalgia
- C. Happiness
- D. Lust

Q.33 Goethe said, "Doubt grows with knowledge". From this perspective, who would a knowledgeable person be?

- A. One for whom doubt does not grow with knowledge
- B. One who has no doubt
- ☒ C. One who has doubt
- ☒ D. One whose doubt grows with knowledge

Q.34 Theory is already existing knowledge and Practice is knowledge that comes by 'doing'.

Based on this statement which of the following is/are TRUE?

- ☒ A. Theory precedes Practice.
- ☒ B. Practice is knowledge in action.
- ☒ C. Knowledge without 'doing' is Theory.
- D. Theory and Practice are one and the same.

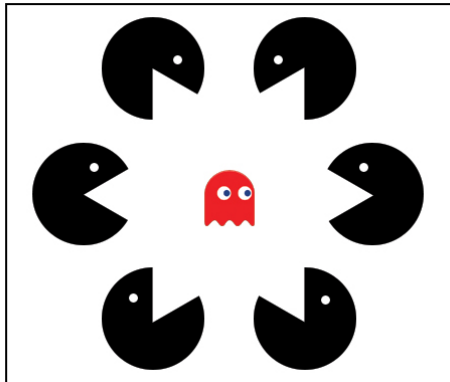
Q.35 Which of the following quotes celebrate(s) technology?

- A. "Did that search engine really know what you want, or are you playing along, lowering your standards to make it seem clever?"
- B. "I fear that we are beginning to design ourselves to suit digital models of us, and I worry about a leaching (leakage) of empathy and humanity in the process"
- C. "The approach to digital culture I abhor would indeed turn all the world's book into one book".
- ☒ D. "What a computer is to me is the most remarkable tool that we can ever come up with. It's the equivalent of a bicycle for our minds".

Q.36 Consider the drip irrigation system in villages on hilly terrain. Which of the following statements is/are correct?

- ☒ A. Bamboo pipes can be used to divert perennial springs on the hilltops to the lower reaches by gravity.
- ☒ B. If the bamboo pipes have to cross a road they can be taken above the land.
- ☐ C. This irrigation system is useful for the places with high ground water level.
- ☒ D. The bamboo pipes could be replaced with PVC pipes.

Q.37 Six pacmen have joined hands to take revenge from an evil ghost. They are standing around in a circular formation. Lo and behold, a white star full of pure light emerges in the centre. In complete disbelief, the pacmen are mesmerized. Which of the gestalt laws of perception explain(s) the appearance of the star in the centre?



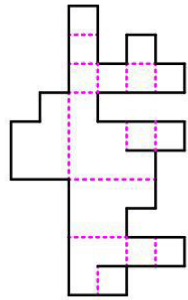
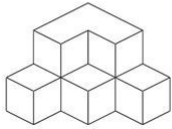
- ☒ A. Proximity
- ☒ B. Continuity
- ☐ C. Figure and ground
- ☒ D. Closure

DESIGNING WITH GESTALT PRINCIPLES

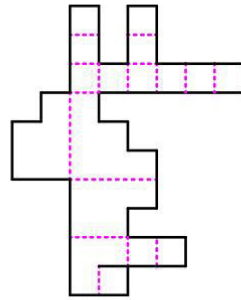
10 primary principles underpin the practical uses of Gestalt Psychology

1 Simplicity Combining simplicity with creativity can lead to stunning creations. How to Master Simplicity: Know how to balance simple shapes with visual stimulation. Give the eye a comfortable form that helps it interpret what it sees.	2 Figure-Ground People can immediately identify which element is the figure, and which is the ground. Use these two related principles to make the most of the figure-ground effect: • Area - The viewer's mind sees the smallest element as the figure and the larger one as the ground or background. • Convexity - Convex elements are related to figures.	3 Proximity Elements close to each other are perceived as part of the same group. Common Use Case: Kerning. Proper kerning helps readers snap up each word.	4 Similarity Elements that look alike are perceived as part of a group. The principle of similarity applies to: • Color • Size • Shape • Texture • Orientation
5 Common Fate Objects that seem to be moving in the same direction are often seen as a group.	6 Symmetry The principle of symmetry applies to: • Mirrored shapes • Balanced elements • Parallel lines	7 Continuity Objects that are plotted in a continuous pattern are grouped together by the mind. Smooth lines often make a unified figure.	8 Closure The mind wants closure. A shape only needs to be implied for the mind to "fill in the gaps" and see what it wants to see.
9 Common Region Elements in one region are seen as one group. Badge designs are seen as unified, despite some containing text, banners, and other objects.	10 Connectedness When there is a connection between elements, it is easy to see a unified figure. Achieve connections with: • Arrow • Lines • Illustrations (i.e. a rope)	Gestalt Theory Humans are bombarded with countless signals day-to-day. To keep from going crazy, we unify these signals into groups. Gestalt designers are obsessed with how people put objects together in their minds. Good designs leads people to experience the message you want to convey. Arm yourself with the principles of Gestalt and create designs that have the impact you want.	

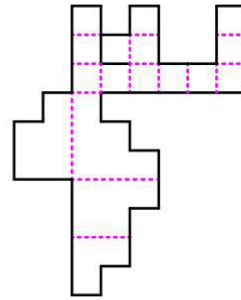
Q.38 If the black lines represent cutting lines and dotted pink lines are folding lines, which of the options correctly represent(s) the opened up surfaces of the solid shown?



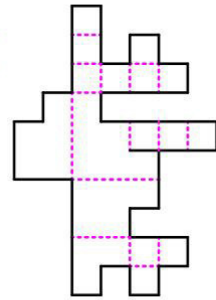
A



~~**B**~~



~~**C**~~



D

Q.39 Which of the following images belong(s) to Ajanta?



~~**H**~~



~~**I**~~



~~**J**~~



~~**K**~~



L



~~**M**~~

- ~~A. H~~
- ~~B. L~~
- ~~C. K and M~~
- ~~D. J and I~~

Q.40 Based on the paintings below, which of the following statements is/are TRUE?



H



I



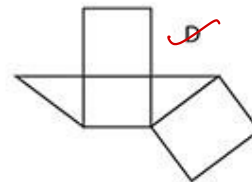
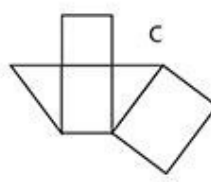
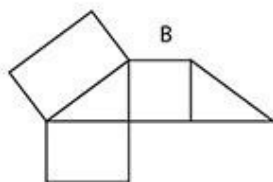
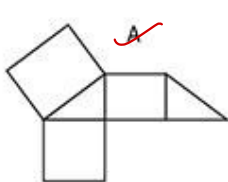
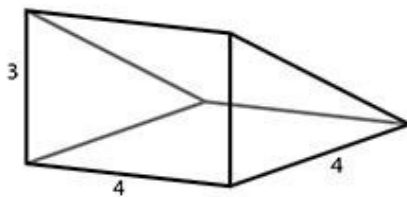
J

- A. These paintings use geometric grids for their construction.
- B. All three painting are from Maharashtra.
- ☒ C. All these paintings draw upon tribal folklore.
- ☒ D. These paintings are wall murals.

Q.41 Which of the following is/are greenhouse gas(es)?

- A. Hydrogen Sulphide
- ☒ B. Carbon Dioxide
- C. Chlorine
- ☒ D. Methane

Q.42 Which of the options represent(s) correctly the opened up surfaces of the solid shown?



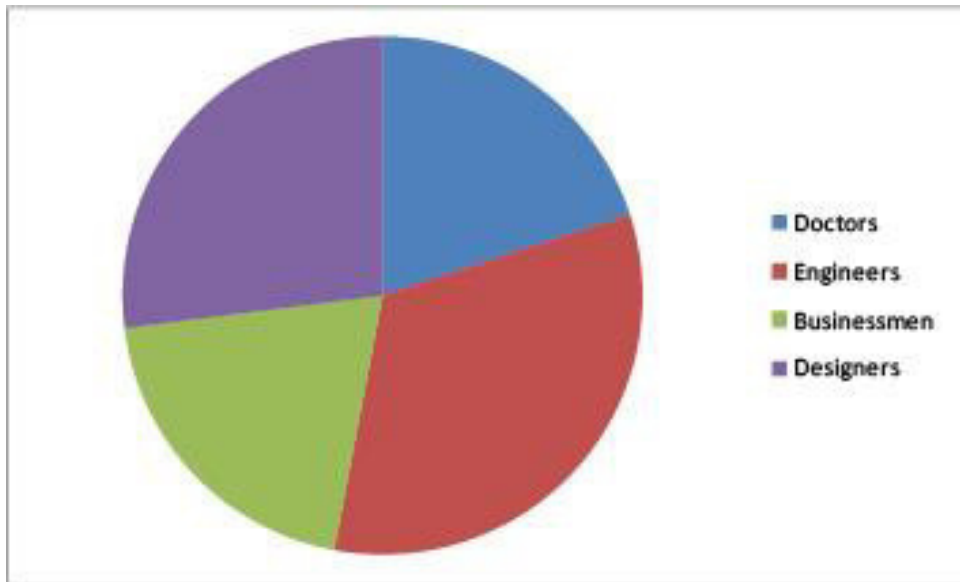
Q.43 Consider the following text by Edward Shanken:

“By 2000, it had become increasingly apparent that the exclusion and ghettoization confronting the practice and criticism of new media art and the larger historiography of art and technology required an explicit suturing strategy. In ‘Art in the Information Age’ (2001) I argued that by ‘interpreting conceptual art and art-and-technology as reflections and constituents of broad cultural transformations during the information age’ categorical distinctions can be relaxed, allowing parallels to be drawn between seemingly diverse theory of systems aesthetics and the notion of software as a metaphor for art. My analysis of works by Levine, Haacke and Kosuth in ‘Software’ led to the conclusion that in the information age, ‘meaning and value are not embedded in objects, institutions or individuals so much as they are abstracted in the production, manipulation and distribution of signs and information’. Finally, I implicitly applied Burnham’s systems approach to analyse the system by which art history is written. Using Haacke and Ascott as examples, I claimed that the historicization of an artist’s work as conceptual art or art and technology ‘says less about their work than it does about the institutional mechanisms that have created and reinforced categorical distinctions...at the expense of identifying continuities between them.’”

Which of the statements is/are TRUE of the paragraph above?

- A. Historicization of an artist’s work speaks only of institutional mechanisms that create and define categories.
- ☒ B. Categorical distinctions do not allow parallels to be drawn between seemingly diverse practices.
- ☒ C. If software is a metaphor for art, meaning and value are more in the making and distribution of signs and information.
- D. Conceptual art cannot be interpreted as reflections of broad cultural transformations.

Q.44 The following diagram shows the distribution of population in an apartment building as per the profession of its residents.

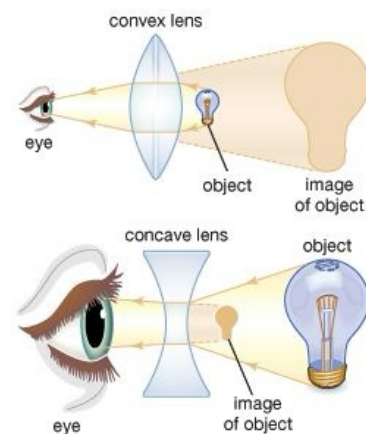


Which of the following is/are correct?

- ✓ A. Doctors and businessmen combined is less than half of the total residents.
- ✓ B. About 1/3 of the population is engineers.
- ✓ C. Engineers and designers combined is about 60% of the residents.
- D. Doctors and businessmen combined is more than half of the total residents.

Q.45 In which of the following applications is a convex lens used?

- ✓ A. Peepholes
- ✓ B. Magnifying glass
- ✓ C. Microscope
- ✓ D. Car headlight

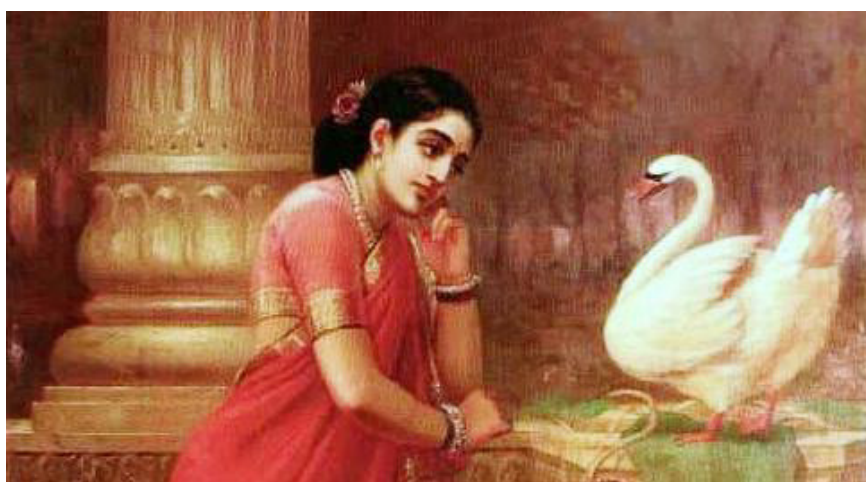


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Section C: Multiple Choice Questions

Section C (120 Marks) contains a total of 40 **Multiple Choice Questions (MCQ)**. Each question has four choices out of which **ONLY ONE** is the correct answer. There is **NEGATIVE** marking for this section. Each correct answer will be awarded 3 marks and each wrong answer will receive -1 (minus 1) mark. Questions not attempted will be given zero marks. Questions from 46 to 85 belong to this section.

Q.46 Who is the woman in this painting by Raja Ravi Varma?



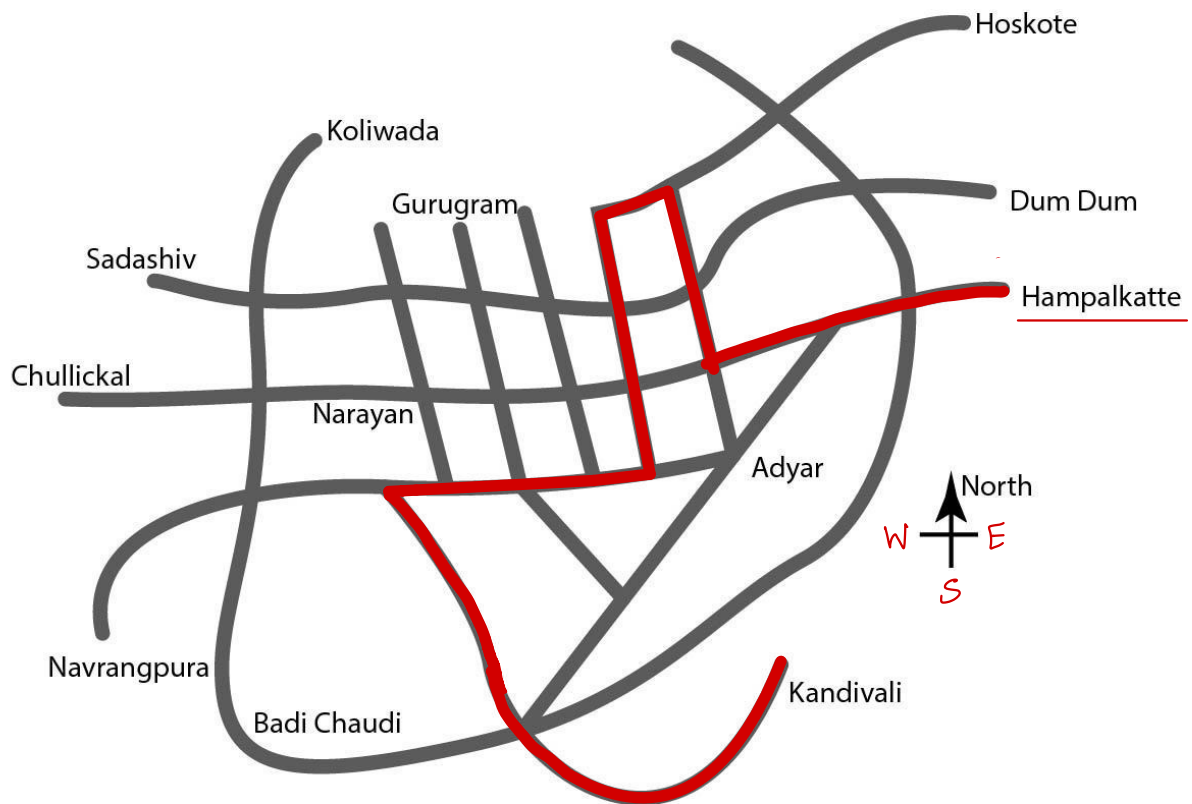
- A. Shakuntala
- B. Lakshmi
- C. Saraswati
- ☒ D. Damayanti

Q.47 Which option arranges the states in an ascending order in terms of area?

- A. Gujarat, Kerala, Uttar Pradesh, Rajasthan
- B. Gujarat, Kerala, Rajasthan, Uttar Pradesh
- ☒ C. Kerala, Gujarat, Uttar Pradesh, Rajasthan
- D. Kerala, Gujarat, Rajasthan, Uttar Pradesh

Sr No	Area wise States	Population wise States
1	Rajasthan	Uttar Pradesh
2	Madhya Pradesh	Maharashtra
3	Maharashtra	Bihar
4	Uttar Pradesh	West Bengal
5	Gujarat	Madhya Pradesh
6	Karnataka	Tamil Nadu
7	Andhra Pradesh	Rajasthan
8	Odisha	Karnataka
9	Chhattisgarh	Gujarat
10	Tamil Nadu	Andhra Pradesh
11	Telangana	Odisha
12	Jammu Kashmir	Telangana
13	Bihar	Kerala
14	West Bengal	Jharkhand
15	Arunachal Pradesh	Assam
16	Jharkhand	Punjab
17	Assam	Chhattisgarh
18	Himachal Pradesh	Haryana
19	Uttarakhand	Jammu Kashmir
20	Punjab	Uttarakhand
21	Haryana	Himachal Pradesh
22	Kerala	Tripura
23	Meghalaya	Meghalaya
24	Manipur	Manipur
25	Mizoram	Nagaland
26	Nagaland	Goa
27	Tripura	Arunachal Pradesh
28	Sikkim	Mizoram
29	Goa	Sikkim

Q.48 Shown below is a map of a city with names of prominent localities. A man starts walking from Hampalkatte towards the west, takes the second right, second left, first left, third right, second left, and walks on straight to the end of the street. Where does he reach?



- ☒ A. Kandivali
- ☐ B. Gurugram
- ☐ C. Navrangpura
- ☐ D. Hoskote

Ans A

Q.49 The door height of entrance of an Indian Public Hall is to be finalised. Which of the following should be considered to decide the height of the door frame?

- ☒ A. The height of tallest 5 percent of Indian male population.
- ☐ B. The height of tallest 50 percent of Indian female population.
- ☐ C. The average height of tallest 95 percent of Indian male and female population.
- ☐ D. The height of tallest 95 percent of Indian male child population.

→ If tallest 5% can enter through door than every one can enter.

Q.50 Two different views of a sphere are shown below. Select the correct digital image that is mapped on the sphere.



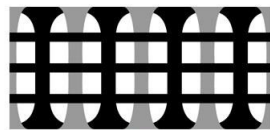
FRONT



BACK



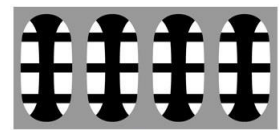
A



B



~~C~~



D

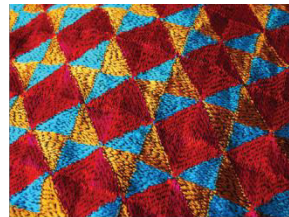
Q.51 Which of these is an example of traditional embroidery from Punjab?



A



B

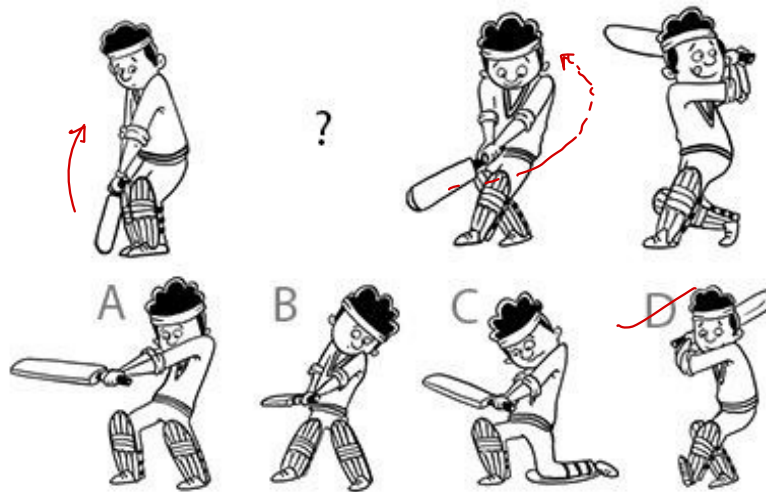


~~C~~



D

Q.52 Given below is a sequence of drawings from the animation of a cricket batsman performing a pull shot, with one missing drawing represented by a question mark. Which option would best replace the question mark?



Q.53 Which option will replace the question mark?

If $\blacklozenge + \blacklozenge = \blacklozenge$
 and $\blacklozenge + \blacktriangle = \blacktriangle$
 and $\blacktriangle + \blacktriangle = \blacktriangle \blacklozenge$
 and $\blacktriangle + \blacktriangle \blacktriangle = \blacktriangle \blacklozenge \blacklozenge$
 and $\blacktriangle + \blacktriangle \blacklozenge = \blacktriangle \blacktriangle$
 then $\blacktriangle \blacklozenge + \blacktriangle \blacklozenge = ?$

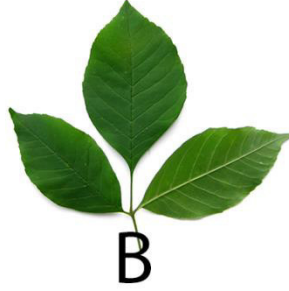
- A $\blacklozenge \blacktriangle \blacktriangle$
 B $\blacklozenge \blacktriangle \blacklozenge$
 C $\blacktriangle \blacklozenge \blacktriangle$
 D $\blacktriangle \blacklozenge \blacklozenge$

Let $\blacklozenge = 0$ and $\blacktriangle = 1$
 The question is in the form of Binary system.

- $\begin{matrix} 0 & + & 0 & = & 0 \\ 2^0 & & 2^0 & & 2^0 \end{matrix}$
- $\begin{matrix} 0 & + & 1 & = & 1 \\ 2^0 & & 2^0 & & 2^0 \end{matrix}$
- $\begin{matrix} 1 & + & 1 & = & 10 \\ 2^0 & & 2^0 & & 2^1 2^0 \end{matrix}$

- $\begin{matrix} 1 & + & 11 & = & 100 \\ 2^0 & & 2^1 2^0 & & 2^2 2^1 2^0 \end{matrix} \quad (\therefore 2^2 \times 1 + 2^1 \times 0 + 2^0 \times 0) = 4$
- $\begin{matrix} 1 & + & 10 & = & 11 \\ 2^0 & & 2^1 2^0 & & 2^1 2^0 \end{matrix} \quad (2^1 \times 1 + 2^0 \times 1) = 3$
- $\begin{matrix} 10 & + & 10 & = & 4 \\ 2^1 2^0 & & 2^1 2^0 & & 2^2 \end{matrix}$
 $(2^1 \times 1 + 2^0 \times 1) = 2$
 $\therefore \blacktriangle \blacklozenge \blacklozenge$

Q.54 Shown below is a flower. Which is the leaf of the plant to which the flower belongs?



55 The word shown below is a specific font. From the options choose the letter that belongs to the same font.

மகிழ்ச்சி

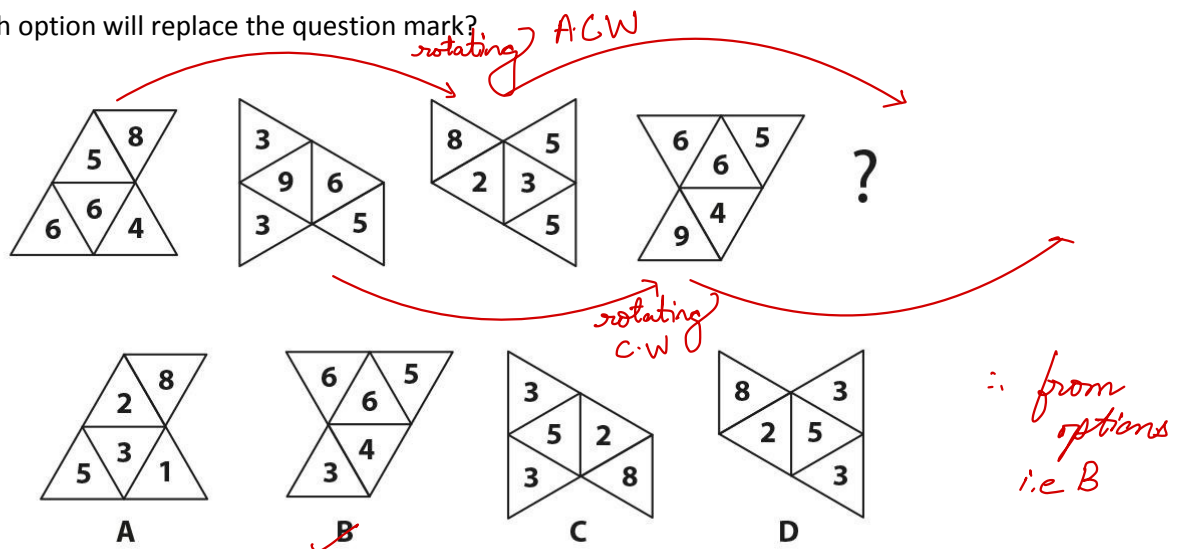
அ
A

ஆ
B

இ
C

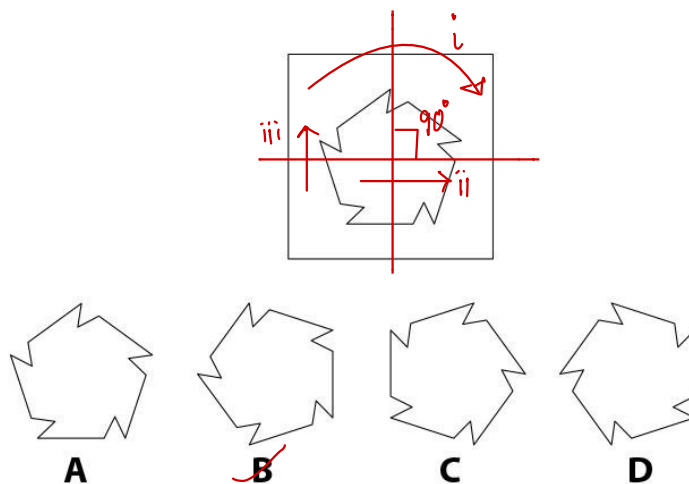
ஊ
D

Q.56 Which option will replace the question mark?



Q.57 If the following operations were performed in the given sequence on the figure shown in the box, what would the resultant figure be?

- Rotate the figure clockwise by 90 degrees.
- Flip along the vertical axis passing through its centre.
- Flip along the horizontal axis passing through its centre.



Q.58 In the left column are names of some National Parks. In the right column are the names of the states in which they are located. Which option contains all correct pairs?

i. Jim Corbett	A. Rajasthan
ii. Ranthambore	B. Assam
iii. Kaziranga	C. Uttarakhand
iv. Kanha	D. Madhya Pradesh

- A. Jim Corbett-Assam; Ranthambore-Rajasthan; Kaziranga-Madhya Pradesh; Kanha-Uttarakhand
- B. Jim Corbett-Uttarakhand; Ranthambore-Madhya Pradesh; Kaziranga-Assam; Kanha-Rajasthan
- C. Jim Corbett-Assam; Ranthambore-Rajasthan; Kaziranga-Uttarakhand; Kanha-Madhya Pradesh
- ☒ D. Jim Corbett-Uttarakhand; Ranthambore-Rajasthan; Kaziranga-Assam; Kanha-Madhya Pradesh

Q.59 Figure 1 is unfolded into Figure 2 and individual letters are written on each face. Which option is the correct representation when it is folded back?

Figure 1

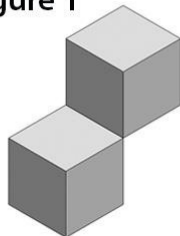
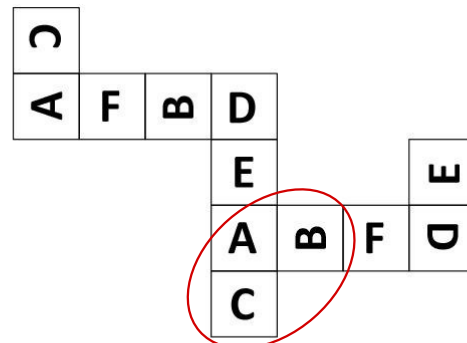


Figure 2



A



B



C



☒ D

Q.60 The image below shows four different dance forms. Which states do these dance forms belong to?

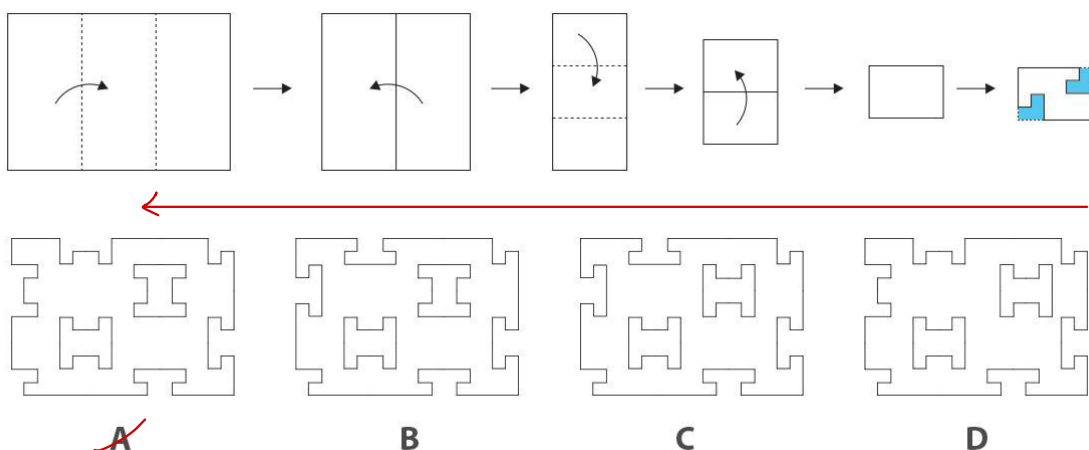


- A. 1-Kerala; 2-Orissa; 3-Andhra Pradesh; 4-Assam
 B. 1-Kerala; 2-Kerala; 3-Andhra Pradesh; 4-Orissa
 C. 1-Kerala; 2-Andhra Pradesh; 3-Kerala; 4-Assam
 D. 1-Orissa; 2-Assam; 3-Kerala; 4-Andhra Pradesh

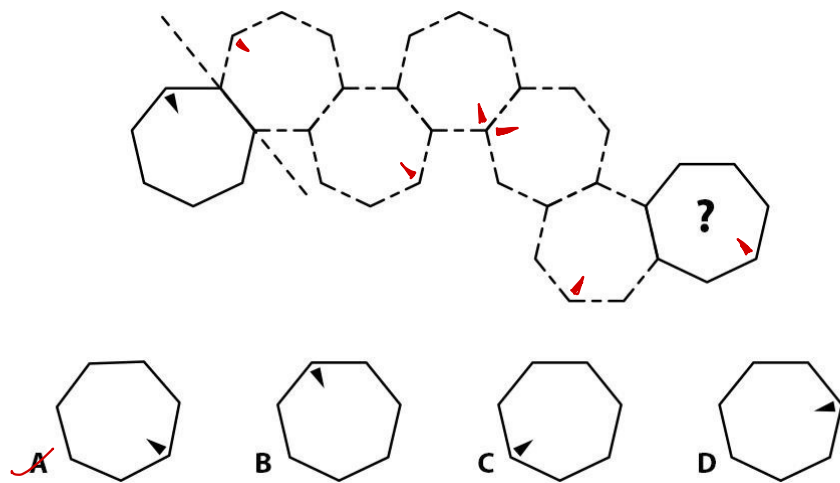
State of Origin	List of Folk Dances in India
Andhra Pradesh	Vasireti Natyam, Bhama Kalapam, Veeranatyam, Dappu, Tappeta Gullu, Lambadi, Dhimesa, Kolattam.
Arunachal Pradesh	Buliyu, Chalo, Wancho, Pasi Kangki, Pomung, Popur
Assam	Bihu, Blichhura, Natpuja, Mahoras, Kalligrapa, Bagunumba, Naga dance, Khep Gopal.
Bihar	Jata-Jatin, Bakho-Bakhain, Panwariya
Chhattisgarh	Gaur Maria, Parthi, Raut Nacha, Pandwani, Vedamati, Kapalik
Gujarat	Garba, Dandiya Raas, Tippiari Jurlun, Bhavai
Goa	Tarangameli, Kol, Dakhni, Fugdi, Shigmo, Ghode, Modhi, Samoyi, Indiya, Jager, Ranmole
Haryana	Jhumar, Phag, Daph, Dhamal, Loor, Gugga, Khor.
Himachal Pradesh	Jhora, Jholi, Chhathi, Dhaman, Chhapeli, Mahasu
Jammu & Kashmir	Raut, Hikot, Mandaps, Kud Dandi Nach
Jharkhand	Akrap, Karma Munda, Agni Jhumar, Jamani Jhumar, Mandana Jhumar, Palika, Phagua
Karnataka	Yakshagana, Huttari, Suggi, Kunitha, Karga
Kerala	Ottam Thullali, Kakkattikali

Maharashtra	Lavani, Nakata, Kol, Leim, Gafa, Dahikola Dasavtar
Madhya Pradesh	Jawara, Marki, Aada, Khada Nach, Phulpoti, Grida Dance, Selatarki, Selabhadani
Manipur	Dol Chalam, Thang Ta, Lal Haraoba, Pung Chalam
Meghalaya	Ka Shad Suk Mysiern, Nongkrem, Laho
Mizoram	Cheraw Dance, Khualam, Chalam, Sawlakin, Chawnglatawn, Zangtalaw
Nagaland	Rangma, Zeliang, Nsuliolians, Gethingim
Odisha	Savari, Ohumara, Palka, Munari
Punjab	Bhangra, Giddha, Daff, Dhaman, Bhand
Rajasthan	Ghumar, Chakri, Ganagor, Jhulan Leela, Jhuma, Suisini, Ghapal
Sikkim	Chu Foot, Sikmar, Singhi Chham or the Snow Lion, Yak Chham, Denzong Gnenha, Tashi Yangku
Tamil Nadu	Kumi, Kolattam, Kavadi
Tripura	Hojagiri
Uttar Pradesh	Nautanki, Rasika, Kajri, Jhora, Chappell
Uttarakhand	Garhwali, Kumauni, Kajari, Jhora, Rasika

Q.61 A paper is folded in the sequence shown in the image below. Blue patches are the areas that are cut out. What would the resultant figure be when the paper is unfolded?

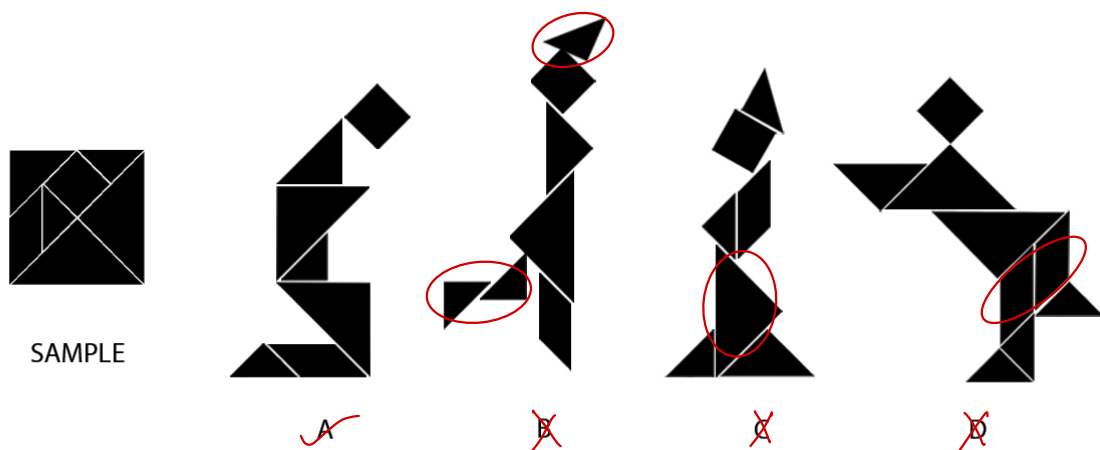


Q.62 If the heptagon with the triangle is mirrored along the common edge (depicted by a dotted line) between two consecutive shapes, what would the resultant figure be?



Ans A

Q.63 Which of the following compositions is possible by rearranging pieces of Tangram puzzle (Sample given on the left)? Rotation of the pieces is allowed but flipping and scaling are not allowed.



- ☒ A) A
- ☐ B) B
- ☐ C) C
- ☐ D) D

Ans A

Q.64 Consider the poem below:

Drip drip Drip Drip
Dripdripdripdrip
Painters far away
Fishermen in oceans of colour

Who or what from the options, drip?

- ☒ A. Painters
- ☒ B. Fishermen
- C. Oceans
- D. Beaches

Q.65 Which of the following crawl on a surface?

- A. Helicopters
- ☒ B. Words
- C. Fruits
- D. Satellites

Q.66 Read the following lines from a poem:

Lad Bazaar *gallis*. Thin and crooked.
On the prime prized precincts
Sat sleeves rolled Ali
Sipping sweet *suleimani*
Props ready, goods ready
Theatre of give and take

What profession could be attributed to Ali?

- ☒ A. Shopkeeper
- B. Teacher
- C. Pilot
- D. Traffic police

Q.67 Read the following lines and answer the question given below:

Pleading farts and ponderous burps
Upset the silence of dignified insides,
Sounds of occupation and exile.

What makes up dignified insides?

- ☒ A. Stomach
- B. Country
- C. Silence
- D. Circus

Q.68 The hour hand of a particular clock moves at half the speed, and the minute hand at 6 times the speed of a regular clock. If this strange clock matches time with the regular clock at 12 noon,
what is the actual time when the strange clock next looks as shown below?



- A. 8.45
- B. 9.05
- ☒ C. 8.05
- D. 10.15

$\therefore$ hour hand has covered 4 hrs
and we know that
Strange clock = $\frac{1}{2}$ actual Clock
 $\therefore$ actual clock = 2×4
= 8 hrs.

min hand has covered 30 min
Strange clock = $6 \times$ actual clock
 $\therefore$ actual clock = $\frac{30}{6}$ min = 5 min

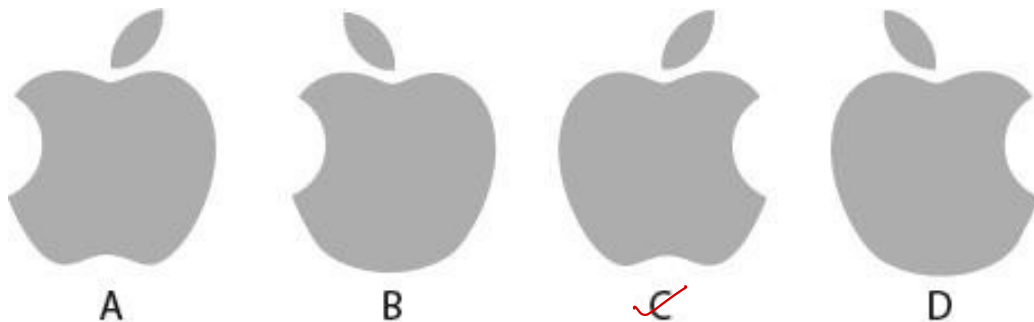
$\therefore$ 8:05

Q.69 Count the number of people with a moustache in the picture below:



- A. 12
- B. 16
- ☒ C. 18
- D. 22

Q.70 Which of the following is the correct logo of Apple Computers?



Q.71 What does a camera use to form an image?

- ☒ A. Convex lens
- B. Concave lens
- C. Both Convex and Concave lens
- D. Plano Concave lens

Q.72 The picture shows a man holding a traditional product in his hands. What is it used for?



- A. Separate rice from husk
- ☒ B. Catch fish in shallow water
- C. Separate tea leaves from dust and foreign particles
- D. Separate gravel from sand

Q.73 Which of the options closely match the Avatars with their associations?

P	Kurma	1	Karma
Q	Parashurama	2	Dharma
R	Rama	3	Complex
S	Krishna	4	Amphibian
T	Kalki	5	Destruction

- A. P-1, Q-2, R-4, S-3, T-5
- B. P-4, Q-5, R-1, S-3, T-2
- C. P-4, Q-5, R-2, S-1, T-3
- D. P-1, Q-3, R-4, S-2, T-5

Q.74 Which of the options correctly match the paintings with corresponding art movements?



H



I



J



K



L



M

1.Expressionism 2.Realism 3.Pop Art 4.Pointilism 5.Op Art 6.Surrealism

A. 1 – I; 2 – H; 3 – K; 4 – J; 5 – L; 6 – M

B. 1 – J; 2 – K; 3 – M; 4 – I; 5 – H; 6 – L

☒ C. 1 – L; 2 – J; 3 – I; 4 – K; 5 – M; 6 – H

D. 1 – H; 2 – K; 3 – M; 4 – I; 5 – L; 6 – J

Q.75 If the following self portraits by Picasso are arranged in chronological order starting with the earliest painting, which one of these options is correct?



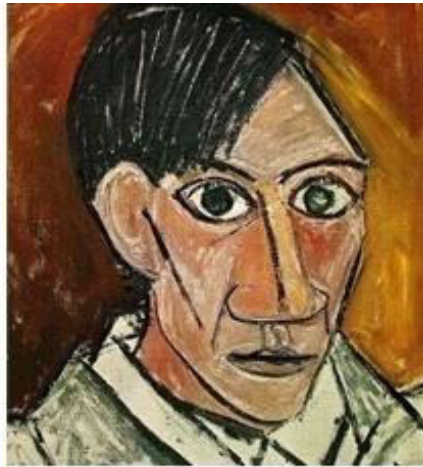
A



B



C



D

- A. B, C, A, D
- B. A, D, B, C
- ☒ C. C, B, D, A
- D. D, B, A, C

Q.76 Where is this famous painting from?

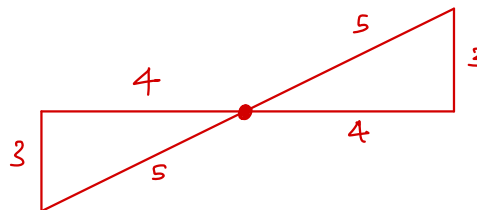


- A. Korea
- B. China
- ☒ C. Japan
- D. Indonesia



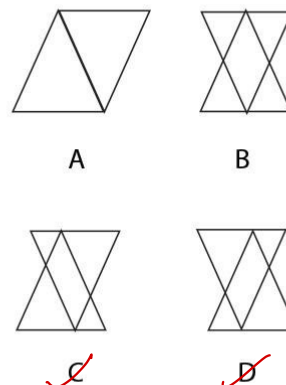
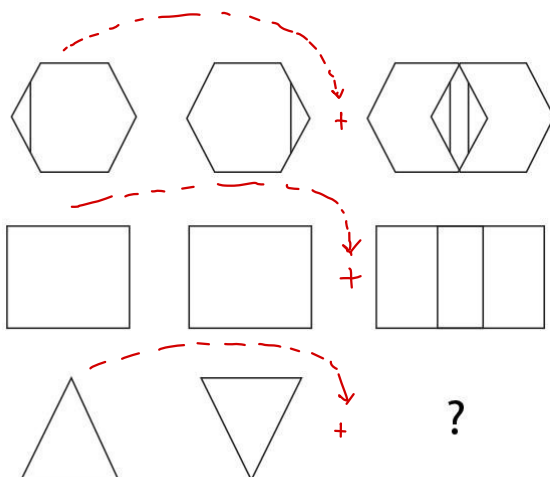
Q.77 Two men start at the same point, walk in opposite directions for 4 meters, turn left and walk another 3 meters. What is the distance between them?

- A. 6 meters
- B. 8 meters
- ☒ C. 10 meters
- D. 12 meters



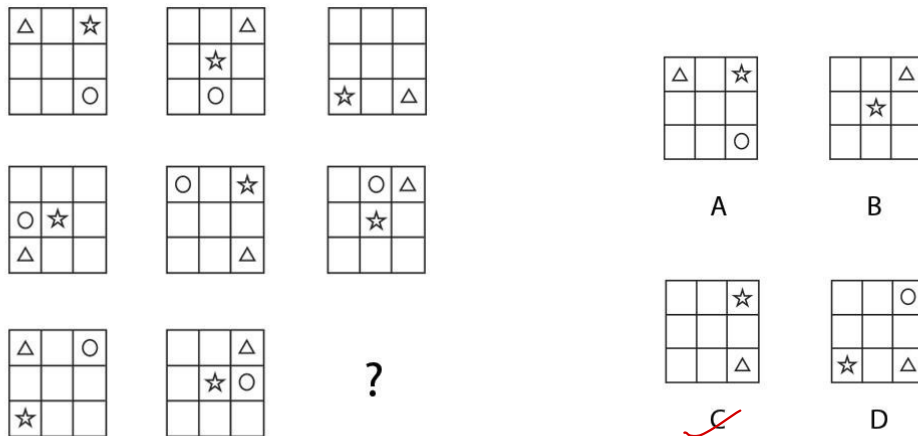
Ans C

Q.78 Which is the appropriate option that can replace the question mark to complete the series?



Ans C, D

Q.79 Which is the appropriate option that can replace the question mark to complete the series?



*Star is moving diagonal
 Δ and O are moving clockwise*

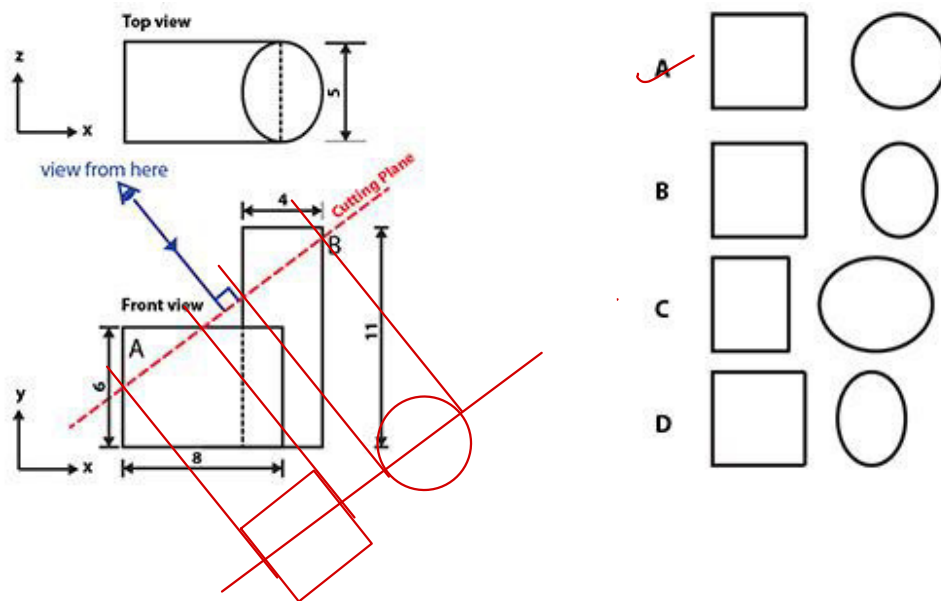
Q.80 Which of the following options is related to the phrase 'Easy to understand'?

- A. Functionality
- ☒ B. Usability
- C. Durability
- D. Aesthetics

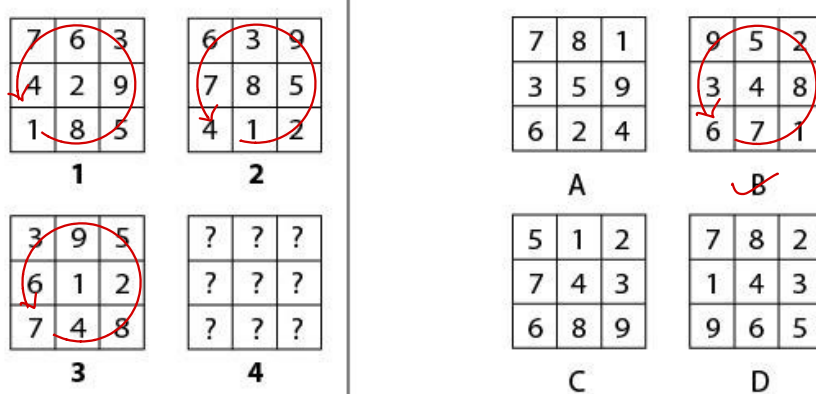
Q.81 Which of the following options helps in establishing interchangeability of products?

- ☒ A. Standardization
- B. Stylization
- C. Diversification
- D. All of the above

Q.82 The figure shows a solid formed by the union of two solids. The first is an elliptic cylinder (cylinder with an elliptical cross-section) of shorter width 4 inches, larger width 5 inches, and height of 11 inches, and the second is a cuboid of 8 inches x 5 inches x 6 inches. The centre of the extruded ellipse is aligned with the edge of the cuboid as shown in the figure. If the resultant solid is cut by a plane (shown by the red dotted line), what will be the cross section of the resulting face as viewed from an angle perpendicular to the face?



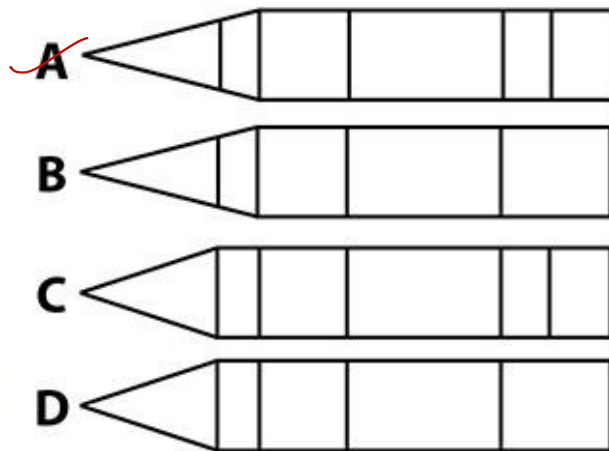
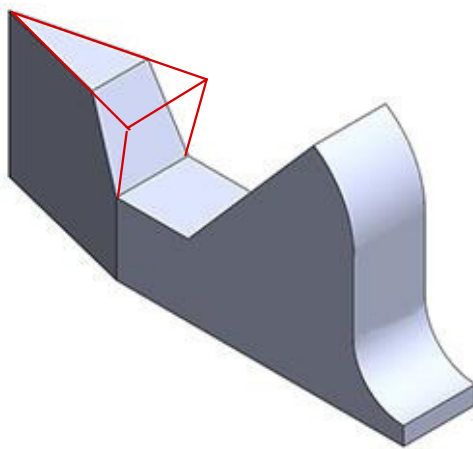
Q.83 A matrix sequence is shown on the left. Which of the options given on the right can replace the matrix with question marks?



Q.84 What does “carbon footprint” refer to?

- ☒ A. The amount of carbon dioxide produced by a certain human activity.
- ☒ B. The total greenhouse gases produced by a certain human activity.
- C. Geographical mapping of the amount of carbon fixed from carbon dioxide by plants through photosynthesis.
- D. The measuring of the presence of prehistoric life based on the fossilised carbon content.

Q.85 Which is the correct top view for the solid shown?



UCEED 2018 FINAL ANSWER KEYS

Type	Q. No.	Answer
NAT	1	21
NAT	2	113
NAT	3	22
NAT	4	12
NAT	5	6
NAT	6	49
NAT	7	7
NAT	8	8
NAT	9	40
NAT	10	46
NAT	11	11
NAT	12	65 or 1.12-1.14
NAT	13	8
NAT	14	25
NAT	15	3
NAT	16	18
NAT	17	20
NAT	18	36
NAT	19	2 or 4
NAT	20	10
MSQ	21	B C D
MSQ	22	B D
MSQ	23	B D
MSQ	24	A B D
MSQ	25	A C
MSQ	26	A B
MSQ	27	A B
MSQ	28	D
MSQ	29	A B C

Type	Q. No.	Answer
MSQ	30	A B
MSQ	31	A B C D
MSQ	32	A B
MSQ	33	C D or D
MSQ	34	A B C or B C
MSQ	35	D
MSQ	36	A B D
MSQ	37	A B D
MSQ	38	B C
MSQ	39	A C D
MSQ	40	C D
MSQ	41	B D
MSQ	42	A D
MSQ	43	B C
MSQ	44	A B C
MSQ	45	B C or A B C or B C D or A B C D
MCQ	46	D
MCQ	47	C
MCQ	48	A
MCQ	49	A
MCQ	50	C
MCQ	51	C
MCQ	52	D
MCQ	53	D
MCQ	54	D
MCQ	55	D
MCQ	56	B
MCQ	57	B

Type	Q. No.	Answer
MCQ	58	D
MCQ	59	D
MCQ	60	C
MCQ	61	A
MCQ	62	A
MCQ	63	A
MCQ	64	A or B
MCQ	65	B
MCQ	66	A
MCQ	67	A
MCQ	68	C
MCQ	69	C
MCQ	70	C
MCQ	71	A
MCQ	72	B
MCQ	73	C
MCQ	74	C
MCQ	75	C
MCQ	76	C
MCQ	77	C
MCQ	78	C or D
MCQ	79	C
MCQ	80	B
MCQ	81	A
MCQ	82	A
MCQ	83	B
MCQ	84	A or B
MCQ	85	A